

02/09/2025

class #14

Electrodinámica

Unit 1:

Electrostatics

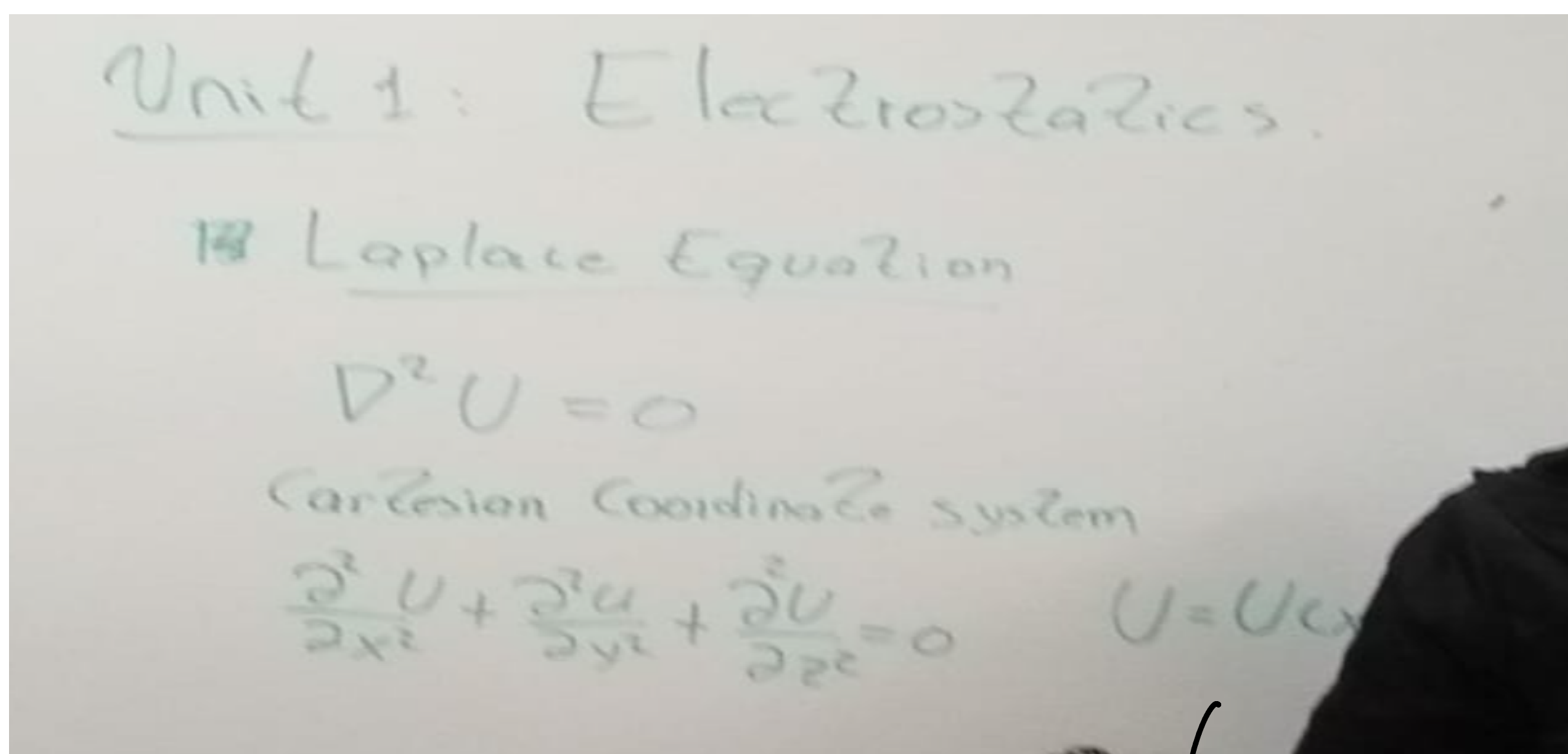
▣ Laplace equation

$$\nabla^2 U = 0$$

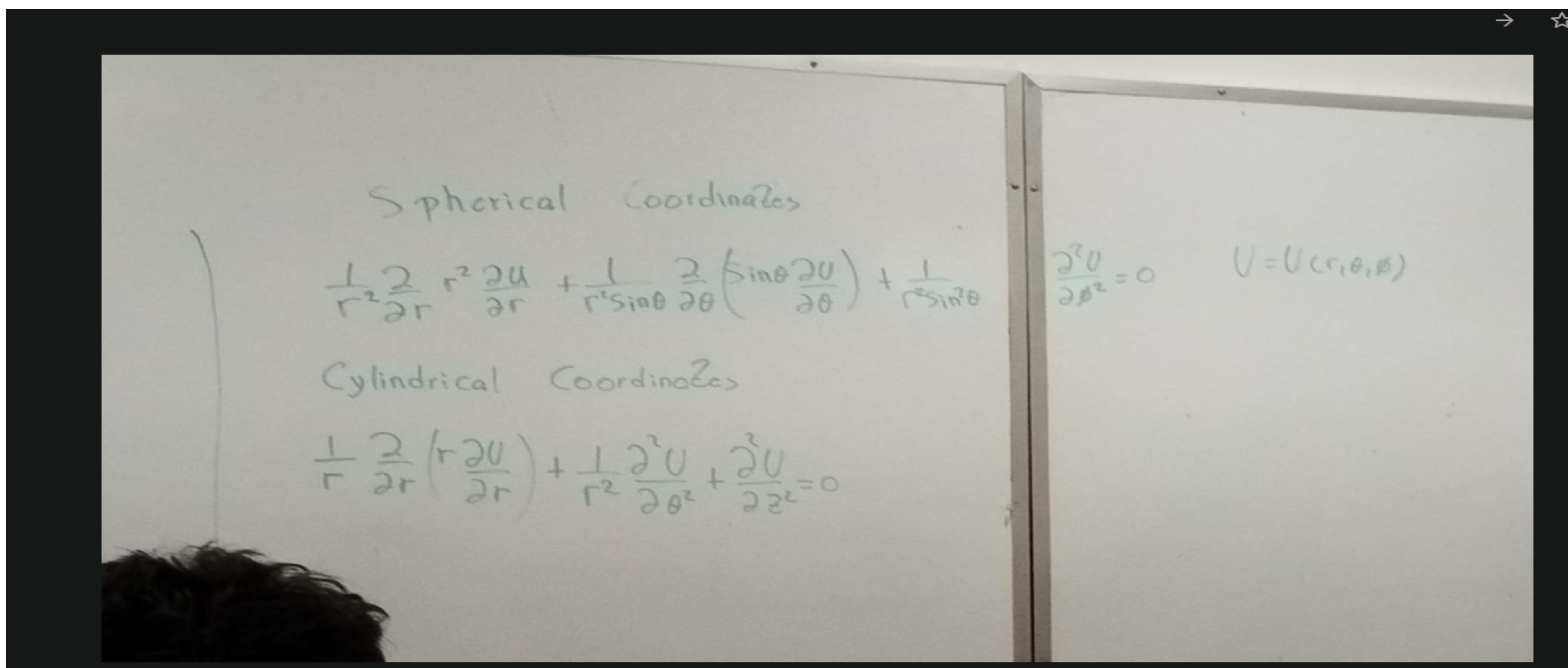
Cartesian

Coordinate system

∂^2



$$U = U(x, y, z)$$



Solutions to Laplace equation

tema de hoy

(a) $U = U(x)$

$$\frac{\partial^2 U}{\partial x^2} = 0 \quad ; \quad \frac{d^2 U}{dx^2} = 0$$

Tenemos

$$U(x) = ax + b$$

b) $U = U(r)$

Handwritten derivation on a whiteboard:

$$U(x) = ax + b$$
$$(b) \quad U = U(r)$$
$$\frac{1}{r^2} \frac{d}{dr} r^2 \frac{dU}{dr} = 0 \Rightarrow \frac{d}{dr} r^2 \frac{dU}{dr} = 0$$
$$\Rightarrow r^2 \frac{dU}{dr} = a$$
$$\Rightarrow \frac{dU}{dr} = \frac{a}{r^2}$$
$$\Rightarrow U(r) = -\frac{a}{r} + b$$

$U(r) = -\frac{a}{r} + b$

Caso c)

Qué pasará si $U = U(r)$;

Handwritten text from a presentation slide:

Unit 1: Electrodynamics. (1)

(c) $U = U(r)$ r cylindrical radial

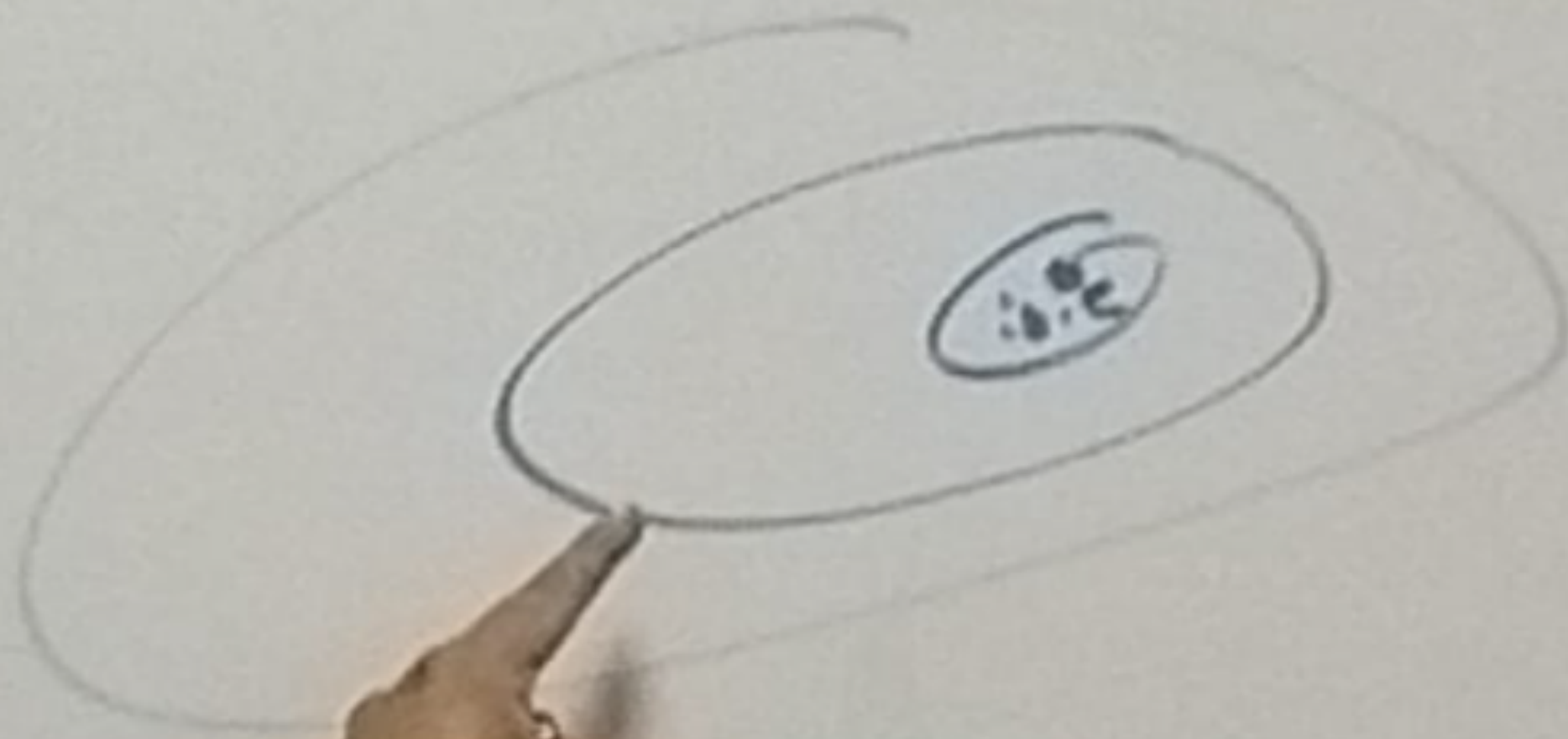
Coordinate.

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dU}{dr} \right) = 0$$

$$\Rightarrow r \frac{dU}{dr} = a$$

$$\Rightarrow \frac{dU}{dr} = \frac{a}{r}$$

$$\Rightarrow U(r) = a \ln r + b$$



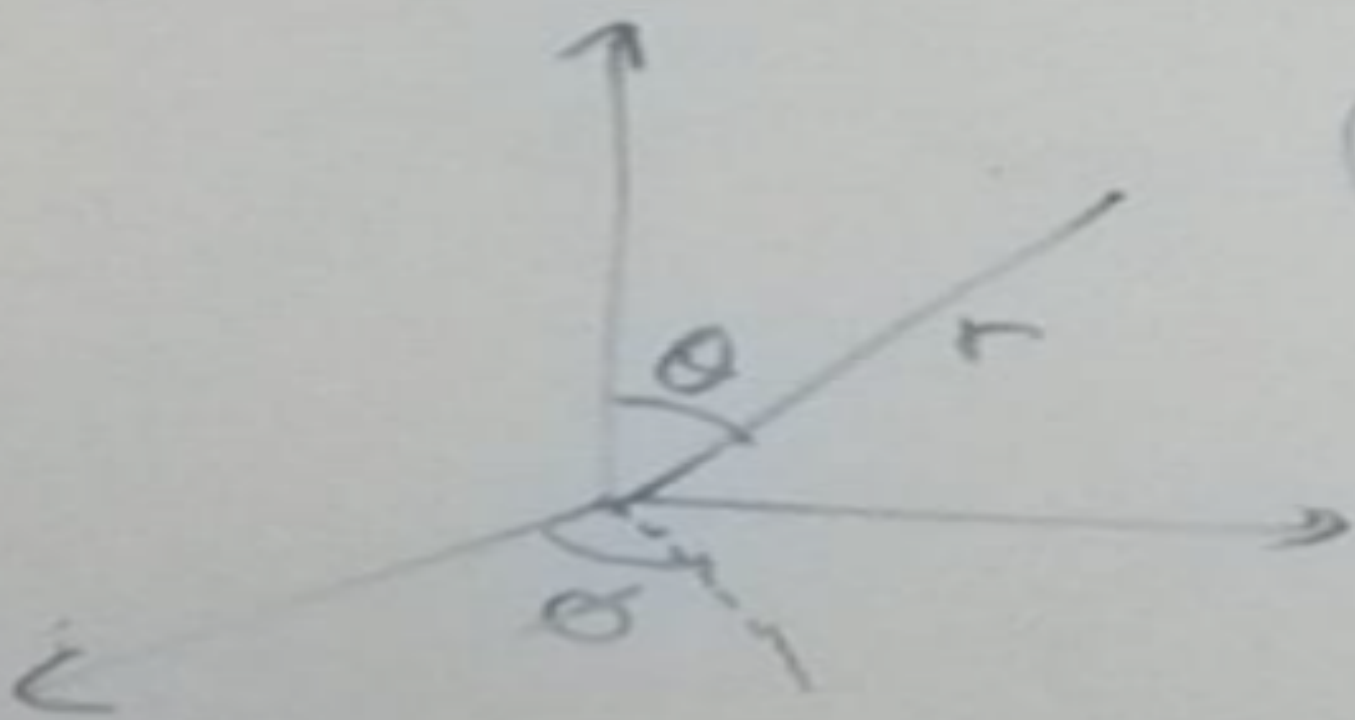
Unit 1: Electro

Classical Electrodynamics

D. Jackson

Última versión del Jackson

☑ Solutions to Laplace equation in spherical coordinates. Zonal Harmonics.



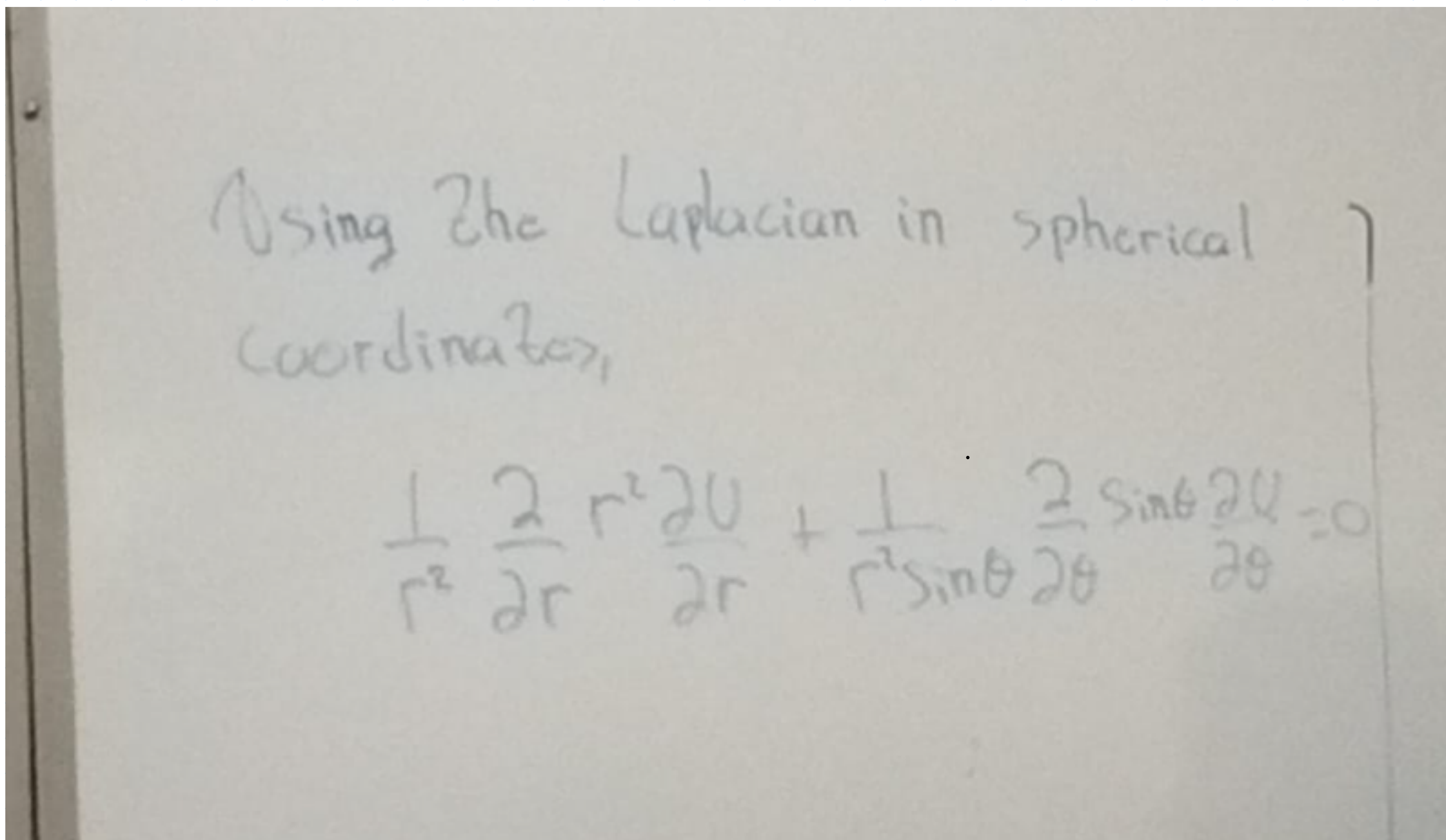
$U(r, \theta, \phi)$

Azimuthal

Symmetry

significa que:

$$U = U(r, \theta)$$



Tenemos

$$U = Z(r) P(\theta)$$

$$\Rightarrow \frac{P}{r^2} \frac{d}{dr} r^2 \frac{dZ}{dr} + \frac{Z}{r^2 \sin \theta} \frac{d}{d\theta} \sin \theta \frac{dP}{d\theta} = 0$$

Tenemos

$$\frac{r^2}{ZP} \left[\frac{P}{r^2} \frac{d}{dr} r^2 \frac{dZ}{dr} + \frac{Z}{r^2 \sin \theta} \frac{d}{d\theta} \sin \theta \frac{dP}{d\theta} \right] = 0$$

Al realizar el producto se tiene:

$$\underbrace{\frac{1}{2} \frac{d}{dr} r^2 \frac{dz}{dr}}_k + \underbrace{\frac{1}{P \sin \theta} \frac{d}{d\theta} \sin \theta \frac{dP}{d\theta}}_{-k} = 0$$

↪ constante

se tiene

$$\frac{1}{P \sin \theta} \frac{d}{d\theta} \sin \theta \frac{dP}{d\theta} + k P = 0$$

} E.C. resultado de la separación propuesta.

E.C. de Legendre

↪ Problema de hacer siglos

} • soluciones que divergen y convergen.

↪ Aproximación mediante serie de potencias (bueno hallar la rel. de recurrencia)

Ecuación de Legendre:

$$(1-x^2) \frac{d^2 P}{dx^2} - 2x \frac{dP}{dx} + k P = 0$$

• Proponer una sol. de la sig. forma:

$$P(x) = \sum_{k=0}^{\infty} a_k x^k \quad \left. \vphantom{\sum_{k=0}^{\infty}} \right\} \text{obtenemos una rel. de recurrencia.}$$

→ si queremos hallar una convergencia la expansión debe
terminar y eso se cumple cuando:

$$\underline{K = n(n+1)}; \quad n \in \mathbb{Z}^+ \longrightarrow \underline{\text{Entero positivo.}}$$

Transformamos $\frac{1}{\sin \theta} \frac{d}{d\theta} \sin \theta \frac{dP}{d\theta} + KP = 0$

A una forma tipo Legendre

↳ $x = \cos \theta$

• Las sol. convergentes reciben el nombre de: Polinomios de Legendre

Polinomios de Legendre evaluados en " $\cos \theta$ ":

$$P_n(\cos \theta)$$

n	$P_n(x)$
0	$P_0(x) = 1$
1	$P_1(x) = x$
2	$P_2(x) = \frac{1}{2}(3x^2 - 1)$
3	$P_3(x) = \frac{1}{2}(5x^3 - 3x)$
4	$P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3)$
5	$P_5(x) = \frac{1}{8}(63x^5 - 70x^3 + 15x)$
6	$P_6(x) = \frac{1}{16}(231x^6 - 315x^4 + 105x^2 - 5)$
7	$P_7(x) = \frac{1}{16}(429x^7 - 693x^5 + 315x^3 - 35x)$

→ Tabla de los pol. de Legendre.

Handwritten notes showing the Legendre differential equation and the first three Legendre polynomials evaluated at $\cos \theta$:

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \sin \theta \frac{dP}{d\theta} + KP = 0$$

$$K = n(n+1)$$

$$P_n(\cos \theta)$$

$$P_0(\cos \theta) = 1$$

$$P_1(\cos \theta) = \cos \theta$$

$$P_2(\cos \theta) = \frac{1}{2}(3\cos^2 \theta - 1)$$

Tenemos

$$\frac{d}{dr} r^2 \frac{dz}{dr} = Kz$$

Assuming that

$$z = r^a$$
$$\Rightarrow \frac{d}{dr} a r^{a+1} = K r^a$$

$$\frac{d}{dr} r^2 \frac{dz}{dr} = Kz$$

Assuming that

$$z = r^a$$
$$\Rightarrow \frac{d}{dr} a r^{a+1} = K r^a$$
$$\Rightarrow a(a+1) r^a = n(n+1) r^a$$
$$\Rightarrow a(a+1) = n(n+1) \quad a = n$$
$$a = -(n+1)$$

The general solution is given by;

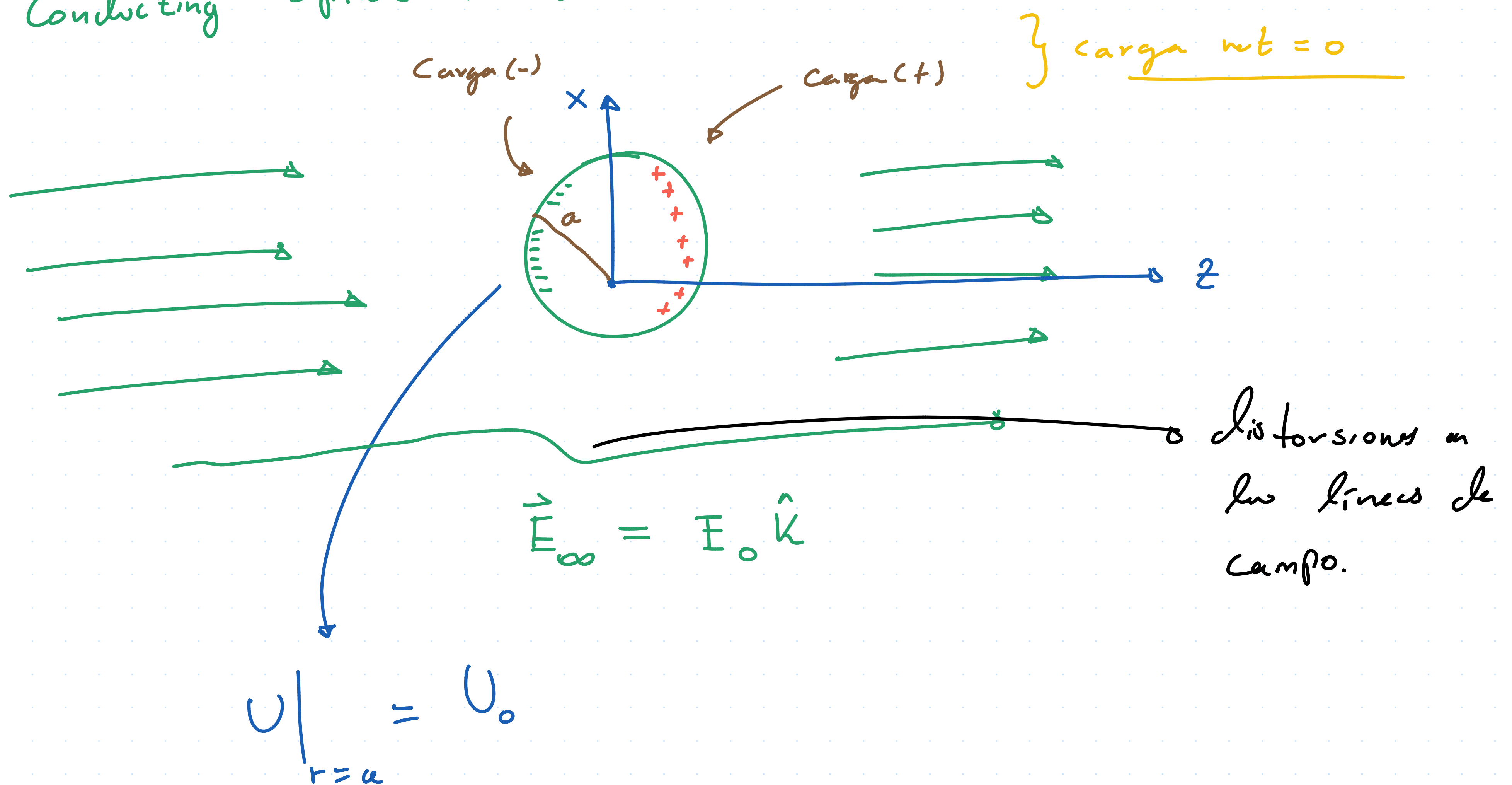
$$U(r, \theta) = \sum_{k=0}^{\infty} A_k r^k P_k(\cos \theta) + C_k r^{-(k+1)} P_k(\cos \theta)$$

Podríamos calcular el campo eléctrico

$$\text{con } \underline{\vec{E} = -\nabla U}$$

Primer ejercicio a resolver:

Conducting sphere in a uniform electric field.



Proponemos la serie:

Para este caso:

$$U = U(r, \theta)$$

Primeros 4 términos:

$$\begin{aligned}
 U(r, \theta) = & A_0 r^0 P_0 \cos \theta + C_0 r^{-1} P_0 \cos \theta + A_1 r^1 P_1 \cos \theta + C_1 r^{-2} P_1 \cos \theta \\
 & + A_2 r^2 P_2 \cos \theta + C_2 r^{-3} P_2 \cos \theta + A_3 r^3 P_3 \cos \theta + C_3 r^{-4} P_3 \cos \theta \\
 & + A_4 r^4 P_4 \cos \theta + C_4 r^{-5} P_5 \cos \theta
 \end{aligned}$$

Tenemos...

$$U(r, \theta) = \sum_{k=0}^{\infty} A_{k+1} r^k P_k(\cos \theta) + C_{k+1} r^{-(k+1)} P_k(\cos \theta)$$

$$U(r, \theta) = A_1 + C_1 r^{-1} + A_2 r \cos \theta + C_2 r^{-2} \cos \theta + A_3 r^2 P_2(\cos \theta) + \\ C_3 r^{-3} P_2 \cos \theta + A_4 r^3 P_3(\cos \theta) + C_4 r^{-4} P_4 \cos \theta + \dots +$$

$$\vec{E}(r, \theta) \Big|_{r \rightarrow \infty} = E_0 \hat{k}$$

$$\vec{E} = -\nabla U$$

$$U(r, \theta) = -E_0 z = -E_0 r \cos \theta$$

$$\Big| U(r, \theta) \Big|_{r=a} = U_0$$

Tenemos la siguiente serie eliminando los términos de orden más grandes...

$$A_i = 0 \quad ; \quad \underline{i \geq 3}$$

Tarea: cómo calcular los coeficientes C_i y qué pasa con términos de orden mayor?

$C = 0$ Para $i \geq 3$

$\hookrightarrow$ vídeo; é por que a partir de C_3 valeu 0 U.

Examen Problemas 2 y 3 Reitz milford